AMFLW-YOLO: A Lightweight Network for Remote Sensing Image Detection Based on Attention Mechanism and Multiscale Feature Fusion

Guili Peng, Zijian Yang, Shoubin Wang, and Yuan Zhou

***Abstract*— The scale of targets in remote sensing images varies greatly and is diverse. It has many small targets that are distributed densely and high complexity of image background. The number of network model parameters and the computation amount of the object detection algorithms based on deep learning is huge. It is difficult to apply them on a platform with fixed performance and limited computing resources. A lightweight remote sensing object detection model is proposed in this article, which called attention and multiscale feature fusion lightweight-YOLO (AMFLW-YOLO). The deep separable convolution, inverted residual, and linear bottleneck structure are employed to replace the standard convolution layer to reduce the model parameters in the backbone network of the model. The coordinate attention (CA) mechanism is introduced into the feature fusion network to capture the direction- and location-aware information across channels at the same time, which improves the accuracy of the network. The bidirectional feature pyramid network (BiFPN) structure is employed to strengthen feature extraction. The learnable weights are introduced to learn the importance of different input features. The multiscale feature fusion is applied to improve the detection effect. The experimental results show that the algorithm achieves satisfactory performance in terms of efficiency and accuracy and has advantages in detection accuracy and model lightweight.**

***Index Terms*— Attention mechanism, feature fusion, lightweight network, object detection, remote sensing image.**

I. Introduction

RECENT years, the quality and quantity of optical remote sensing images increase substantially with the rapid development of optical remote sensing technology and satellite communication technology [1]. Remote sensing image target detection is a basic headache in this field. Its task is to detect and calibrate effective targets in remote sensing images. It plays a vital role in the fields of national defense, military, urban planning, traffic management, and resource exploration. Object detection tasks in optical remote sensing images are similar to object detection tasks in natural scene images. Nevertheless, optical remote sensing images are different from natural scene images in terms of shooting angle, target distribution, and imaging range, bringing more difficulties and challenges to the target detection task of optical remote sensing images [2].

Traditional object detection algorithms usually have high time complexity, and their generalization ability is not strong due to the artificially designed features. Therefore, it cannot accurately identify the category and location information of the target to be detected in the remote sensing image. In recent years, with the rapid development of artificial intelligence (AI) technology and computer, object detection algorithms based on deep learning, especially convolutional neural networks (CNNs), have attracted great research attention. Due to the powerful feature extraction ability of CNN model [3], researchers have developed a large number of algorithms for optical remote sensing image classification and object detection based on CNN, which can be trained and independently learned on GPU to obtain strong robust target features. Compared with traditional target detection algorithms, the detection accuracy has been significantly improved.

Remote sensing has a large variable target scale, uneven distribution, and many dense small targets [4]. Under the same resolution, remote sensing images contain targets of different scales, and the targets of different categories are quite different [5]. There are also large differences between targets of the same category and different subcategories, the distribution of target types is uneven, and the distribution of targets in some areas is relatively sparse [6]. However, the targets in some areas are highly clustered and most of them are small-sized targets. Although the sensory field of the network gradually becomes larger as the number of network layers deepens and the semantic representation of the CNN model is then dramatically enhanced, this also makes the resolution of the image lower, and many detailed features become more and more blurred after the convolutional operation of the multilayer network. The shallow layer network has strong ability to represent geometric detail information, but the semantic nature of the extracted features is weak. In order to obtain strong semantic features, the detector represented by

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Guili Peng, Zijian Yang, and Shoubin Wang are with the School of Control and Mechanical Engineering, Tianjin Chengjian University, Tianjin 300384, China (e-mail: penguili@126.com; yzj19990512@163.com; wsb800@126.com).

Yuan Zhou is with the School of Instrument Science and Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: yuanzhou277@163.com).

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R-CNN [7] in the early stage directly predicted on the last layer feature map of the neural network, but due to the lack of fine-grained space features, its detection effect on small targets was poor. Standard CNNs with output single feature map cannot meet the needs of remote sensing target detection at present [8]. The inability to effectively represent multiscale objects simultaneously using a single layer of feature maps is called the multiscale problem. In order to seek multiscale feature representation, Wang et al. [9] used residual-connected autoencoder multiscale structure to adapt to changes in different scales. Hou et al. [10] adopted multiple parallel ROI Pooling modules to map features to different scales and a multistage cascade method for detection. Based on the existing Faster R-CNN [11], Zhang et al. [12] introduced different candidate region extraction networks in parallel on different feature maps to independently detect targets at different scales. Singh and Davis [13] proposed a training strategy called scale normalized image pyramid (SNIP) to address the challenges posed by large numbers of small targets in MS COCO datasets, this strategy can be applied to the two stages of Faster R-CNN at the same time and can improve the detection accuracy of all scales.

Due to the limitation of single-scale features, multiscale feature fusion has been paid more and more attention by academia and industry [14]. In particular, multiscale feature fusion has been proved to be effective in multiscale target detection in remote sensing images. In the field of remote sensing target detection, recent studies have extensively built their methods on this idea [15], [16], [17] so that features of different resolutions have strong semantic information and rich location information at the same time. Yang et al. [18] adopted a dense feature pyramid network to further strengthen the relationship between features of different scales. Zhang et al. [19] presented a context-aware detection network (CAD-Net) that exploits attention-modulated features as well as global and local contexts to address the problem of multiscale detection of remote sensing images. Fu et al. [20] proposed a feature fusion architecture to generate a multiscale feature hierarchy, which augments the features of shallow layers with semantic representations via a top-down pathway and combines the feature maps of top layers with low-level information by a bottom-up pathway to further enhance the feature representation capability. Based on multiscale detection, Yan et al. [21] balanced the training weights of different scale targets for the loss function to enhance the robustness of the algorithm to different scales. Combining the advantages of CNN and graph convolutional network (GCN), Zhou et al. [22] proposed a novel multiple feature fusion model termed attention multihop graph and multiscale convolutional fusion network (AMGCFN), which included two subnetworks of multiscale fully CNN and multihop GCN to extract the multilevel information of hyperspectral image. Wang et al. [23], [24] successively proposed representation-enhanced states replay network and multistage self-guided separation network, respectively. The former adopted a novel cross-modal interactive fusion method to jointly optimize the parameters of different branches by combining multisource information, which improved the interactivity of modal fusion. The latter utilized a self-guided network to find common features between intraclass samples and improve the consistency of feature representation. It combined the texture and morphological features of images to guide feature learning, effectively reducing the impact of intraclass differences. For the multimodal remote sensing data classification task and infrared image small target detection task, Wu et al. [25], [26] successively proposed the cross-channel reconstruction module CCR-Net and UIU-Net, respectively. The former learns more compact fusion representations of different remote sensing data sources by means of the reconstruction strategy across modalities. The latter integrates Residual U-blocks into a deep supervision network to achieve multilevel and multiscale representation learning of objects. Yao et al. [27] proposed a novel multimodal deep learning framework ExViT, further performance increase attained by a full token-based decision-level fusion module. Luo et al. [28] proposed a multiscale diff-changed feature fusion network (MSDFFN) for hyperspectral image change detection, which improved the ability of feature representation by learning the refined change components between bitemporal hyperspectral images under different scales and represented the subtle changes between bitemporal HSIs in each scale while considering the multiscale features.

Due to the particularity of remote sensing target detection task, remote sensing image target detection needs to consider not only the accuracy of detection but also the deployment difficulty and operational efficiency of practical engineering applications. Object detection algorithms based on deep learning generally focus on building a deep, high-width neural network and improve detection accuracy by increasing the complexity of the network model. The huge network scale makes it difficult to deploy on platforms with fixed performance and limited computing resources, such as satellites, drones, and other embedded devices. They are not suitable for remote sensing object detection tasks.

Lightweight network technology is the key to addressing the real-time detection of targets by neural networks on hardware platforms with limited computing resources. Its core idea is to design a simplified neural network structure to curtail network parameters and computation. Typical lightweight networks comprise SqueezeNet [29], ShuffleNet [30], MobilenetV3 [31], EspNet [32], EfficientNet [33], and GhostNet [34]. Most of the research on lightweight networks focuses on image classification tasks, and there is less research on lightweight networks for object detection tasks. This is in that the lightweight network structure is too simplistic to fully extract the feature information and location information of the target to be detected. If the network is simply lightweighted to detect remote sensing targets, the detection accuracy is far from satisfactory.

The high precision and lightweight are crucial for the application and deployment of remote sensing image target detection in practical engineering. How to build a high-precision lightweight remote sensing target detection network is worthy of further study. Sharma et al. [35] proposed YOLOrs-lite, which counteracted overparameterization by enforcing tensor-train structure to its convolutional kernels. The network combined light storage with real-time inference, making

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it quite promising for edge deployment. Han et al. [36] proposed a lightweight fully convolution network named LWCDNet for change detection. Huang et al. [37] proposed a direction-adaptive knowledge extraction and distillation (DKED) method. This method effectively improved the performance of lightweight remote sensing target detection model without increasing the inference cost.

In summary, the conflict between model accuracy and computational power as well as storage capacity have been impeding the advancement of object detection algorithms in remote sensing images. The existing approaches typically focus on constructing deep and high-width neural networks and enhancing detection accuracy by increasing the complexity of the model. The deployment difficulty and operation efficiency of actual engineering applications are always ignored. The lightweighting technique could greatly reduce the network parameters and computation. Nevertheless, remote sensing images have the characteristics of diverse scales, small targets, large numbers, high density, and complex imaging backgrounds. The simplified network structure largely restricts the ability of the model to extract the target feature information and the position information, and the loss of accuracy tends to be larger.

In order to remedy the above deficiencies, a lightweight target detection model, attention and multiscale feature fusion lightweight-YOLO (AMFLW-YOLO), based on the YOLOv5 [38] architecture is proposed in this article. A lightweight feature extraction network is constructed based on MobileNet, which is representative among lightweight neural networks. In order to meet the needs of deploying remote sensing object detection algorithms on mobile terminals and embedded devices, we extend the lightweight module in MobileNet to obtain a new feature extraction network to realize the lightweight of the model. Specifically, the standard convolution module in YOLOv5 is replaced with depth-separable convolution and inverted residual linear bottleneck structure. At the same time, the coordinate attention (CA) mechanism is introduced in the feature extraction network to enhance features by emphasizing information representation. In addition, a new feature fusion layer is proposed. In the feature fusion layer, the CA mechanism module and bidirectional feature pyramid network (BiFPN) structure are introduced. CA mechanism is used to consider both channel and location information to improve the network attention to positive samples and avoid introducing a lot of overhead. Two-way weighted feature fusion is used to enhance feature extraction, and learnable weights are introduced to better balance the feature information of different scales, so as to improve the detection ability of objects of different sizes.

The main contributions of this work are summarized as follows.

1) A lightweight feature extraction network module incorporating CA mechanism is proposed. The module employs deeply separable convolution and inverted residual linear bottleneck structures to dramatically reduce the model size and parameters. The CA mechanism is utilized to achieve more efficient feature extraction to meet the real-time remote sensing target detection requirements under the resource-constrained conditions of hardware platforms.
2) In order to compensate for the loss of accuracy caused by model lightweighting, the CA mechanism is introduced in the feature fusion process, which captures cross-channel, direction-aware, and location-aware information at the same time. Combined with lightweight feature extraction, it considers both network efficiency and accuracy, which brings a large gain to the remote sensing target detection task.
3) In order to better cope with the multiscale problem, an innovative use of efficient bidirectional cross-scale connection and weighted feature fusion structure is used to achieve simple and efficient multiscale feature fusion. The contextual information of the original remote sensing image is fully utilized to improve the multiscale feature extraction capability of the network, which improves the detection effect against multiscale targets.

The rest of this article is organized as follows. In Section II, the traditional manual features algorithm and the deep learning detection algorithm are analyzed for the research status, and the algorithm we proposed is introduced. In Section III, the AMFLW-YOLO network proposed in this article is introduced in detail. In Section IV, the experimental device and results are discussed and compared with the other research results.

II. Related Work

Detecting and identifying target information in remote sensing images has always been a hot topic in the field of remote sensing image processing and analysis. In recent decades, research on this issue has been conducted around the world. The current mainstream remote sensing target detection algorithms could be divided into traditional algorithms based on manual features and target detection algorithms based on deep learning.

A. Object Detection Algorithm Based on Manual Features

Kim et al. [39] employed a rectangular template instead of a section map and tracked the urban strip roads in urban remote sensing images through the least squares correlation template matching method. The adaptive binary HMT method proposed by Weber and Lefevre [40] has structural elements of different shapes and sizes and could detect objects such as buildings, coastlines, and oil tanks in remote sensing images. However, rigid template matching requires high template accuracy and is sensitive to shape and density variations. The detection effect is poor when the viewing angle of the target changes or the intraclass difference is large. Bai et al. [41] proposed a new feature descriptor combining scale invariant feature transform (SIFT) and target spatial information and employed support vector machine (SVM) to convert the detection task into a ranking query task to detect the target of interest in remote sensing images. Shi et al. [42] regarded ship targets in remote sensing images as anomalies, extracted candidate ship regions through hyperspectral algorithms, and enhanced HOG features using AdaBoost to generate hypotheses and detect ship targets. The target detection method based on traditional machine

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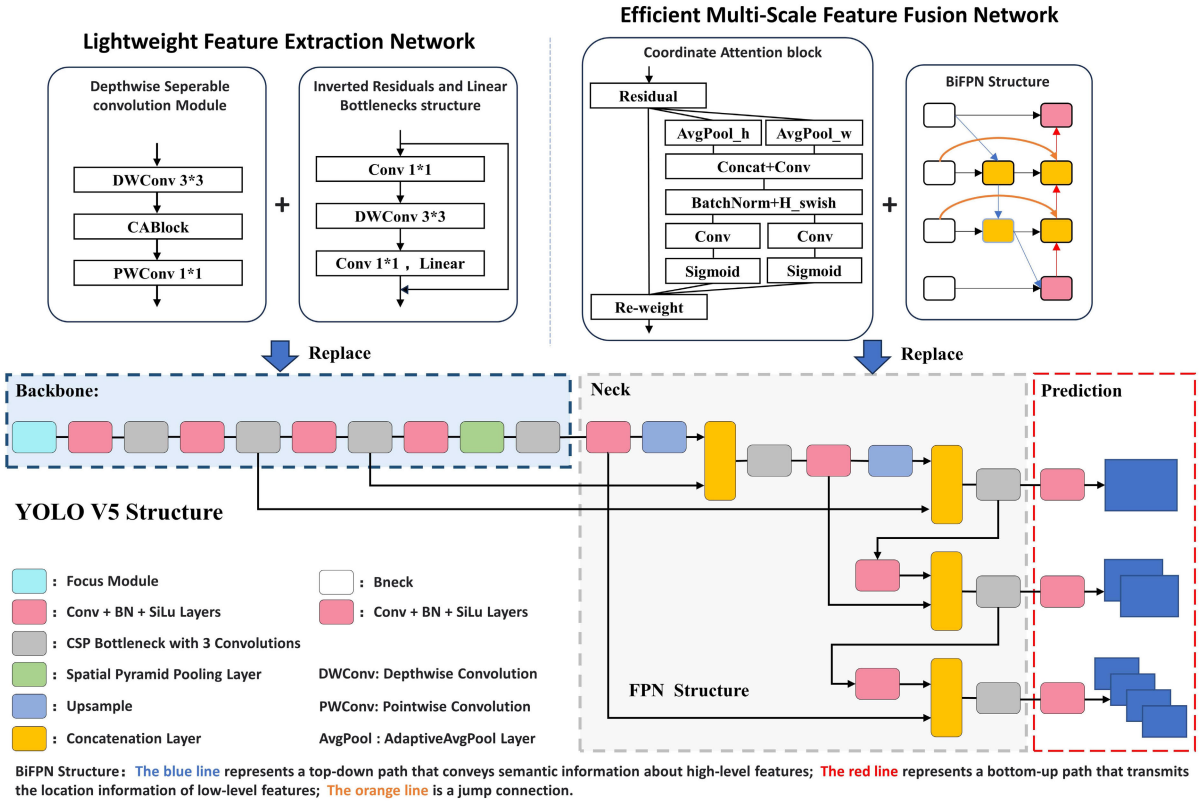


Fig. 1. Optimization techniques used in AMFLW-YOLO. A new feature extraction network is constructed by using depth-separable convolution and inverted residual linear bottleneck structure to replace the original backbone. CA mechanism and BiFPN structure are introduced to construct a new feature fusion network, which replaces the original neck layer.

learning has achieved certain results through the combination of feature extraction and classifier training. However, the features extracted by traditional feature extraction algorithms are essentially low-level features without high-level semantic information, and specific features need to be designed for specific types, making it difficult to effectively represent target features. Concurrently, the performance of the feature classifier is significantly affected by the set parameters and training samples, and the experimental results are not satisfactory.

B. Object Detection Algorithm Based on Deep Learning

In recent years, a large number of researchers in the field of remote sensing image processing have tried to introduce deep learning technology into remote sensing image target detection tasks and conducted a lot of related research. Cheng et al. [43] proposed a novel and effective method for learning a rotation-invariant CNN model to detect remote sensing objects efficiently by introducing a new rotation-invariant layer in the R-CNN framework and optimizing a new objective function. Based on the YOLOv2 network, Van Etten [44] designed a target detection model YOLT for quickly detecting ships, aircraft, and airports in satellite images. Xu and Wu [45] proposed a YOLO-based feature enhancement network for remote sensing target detection. The network model was based on YOLOv3, and a feature enhancement module was added for the difficulties of a large number of small remote sensing targets and complex backgrounds. Li et al. [46] proposed a new HAD baseline network LRR-Net, by combining a low-rank representation model with deep learning techniques to improve the representation performance of hyperspectral anomaly detection. Xie et al. [47] designed a target detection algorithm based on YOLOv4 for remote sensing target detection with dense distribution and complex background. Utilizing the K-means clustering algorithm to cluster the number of anchors, the author reduced the amount of model calculation. The detection of densely distributed objects was achieved by using residual connections rather than continuous convolutions in the original model. Finally, an adaptive feature activation factor was added in the feature extraction stage. Concerning the small target detection problem, a method based on dense connection and multiscale feature fusion was proposed and trained and tested on the Kaggle dataset, so as to enhance the accuracy compared with the original YOLO model.

III. Proposed Method

As shown in Fig. 1, we optimize and improve the YOLOv5-large network and obtain a lightweight and efficient network named AMFLW-YOLO: 1) in order to reduce model parameters and computational load, we utilize two lightweight modules, depthwise separable convolution and inverted residual linear bottleneck structure, to form a new backbone; 2) the CA module is inserted in the neck layer so that the lightweight network could locate and identify the target of interest more accurately and avoid a lot of computing overhead; and 3) we employ the BiFPN structure to replace the FPN in the original

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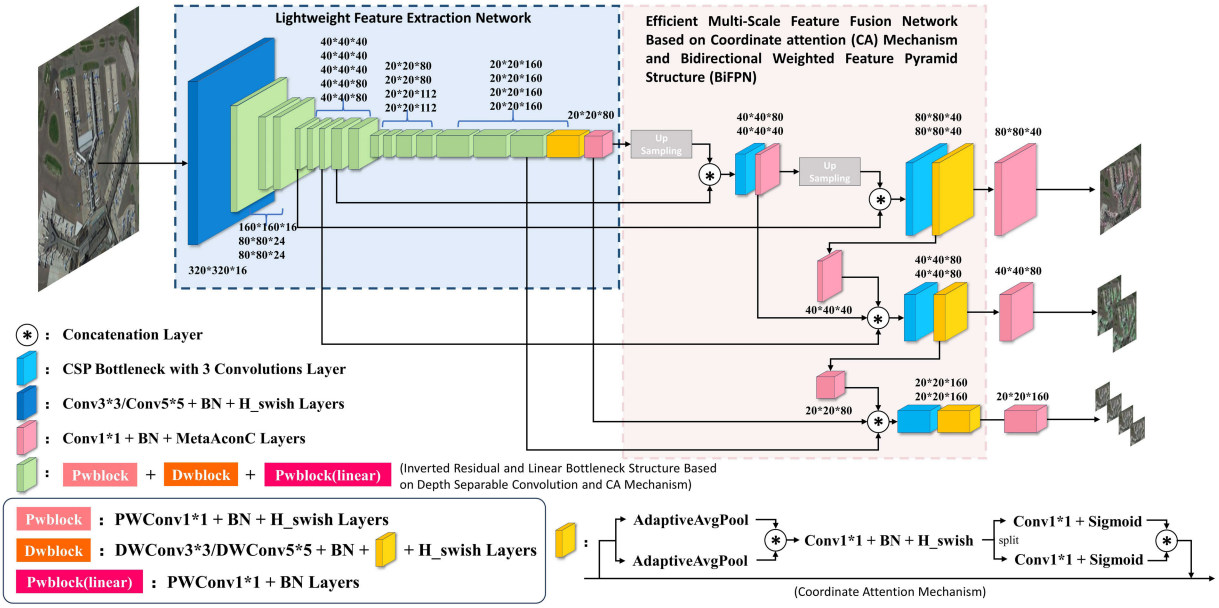


Fig. 2. AMFLW-YOLO network structure.

structure and adopt a more efficient multiscale fusion method to improve the accuracy. In addition, meta-AconC activation function and cosine annealing learning rate strategy are used to train our network, so as to accelerate the convergence of the model and improve the training speed. The AMFLW-YOLO network structure is shown in Fig. 2.

A. Lightweight Feature Extraction Network Module

Compared with processing on the already trained model, the main idea of lightweight model design is to improve the convolution method and design a more efficient network computing method to reinforce detection efficiency. Given a large number of network parameters and computational complexity, the feature extraction network module of the YOLOv5 architecture is optimized in our study. The standard convolution in the original backbone is replaced with the depthwise separable convolution module to build a lightweight feature extractor. In the case of a slight decrease in accuracy, the number of parameters and calculations is significantly lessened. Depthwise separable convolution was proposed in MobileNetV1 [48]. Depth-separable convolution adopts different convolution kernels for different input channels and decomposes the ordinary convolution operation into two processes: channel-by-channel convolution and point-by-point convolution. Each convolution kernel of depthwise convolution is only responsible for one channel. The number of feature maps after completion is the same as the number of channels in the input layer, and the feature maps cannot be expanded. Moreover, the depthwise convolution operation independently performs convolution operations on each channel of the input layer and does not effectively use the feature information of different channels at the same spatial position. Therefore, pointwise convolution is required to combine these feature maps to generate new feature maps. Pointwise convolution is equivalent to a standard convolution with a convolution

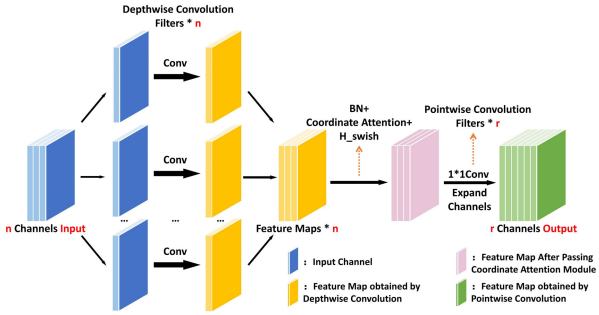


Fig. 3. Depth-separable convolution module structure with CA mechanism.

kernel size of 1×1. The feature maps of the previous step are weighted and combined in the depth direction to generate new feature maps, so as to assure that the depth-separable convolution has the same output dimension as the standard convolution. In our study, a CA mechanism between depthwise convolution and pointwise convolution is inserted to form a new depthwise separable convolution module for strengthening the sensitivity of the network to direction and position information. Its principle will be described later. Fig. 3 shows the improved depthwise separable convolution module structure.

The validity of the lightweight model is verified by calculating the number of parameters and the amount of calculation. The number of parameters for standard convolution and depthwise separable convolution is calculated by the following formulas:

$$N_{\text{std}} = \text{Out}_{\text{ch}} * k^2 * \text{In}_{\text{ch}} \tag{1}$$

$$N_{\text{dw}} = k^2 * \text{In}_{\text{ch}} \tag{2}$$

$$N_{\text{pw}} = 1 * 1 * \text{In}_{\text{ch}} * \text{Out}_{\text{ch}} \tag{3}$$

$$N_{\text{sep}} = N_{\text{dw}} + N_{\text{pw}} \tag{4}$$

where N_{std} denotes the standard convolution parameter quantity, N_{sep} represents the amount of depthwise separable convolution parameters, In_{ch} indicates the number of input

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channels, Out_{ch} refers to the number of output channels, and k signifies the convolution kernel size.

The computational quantity of standard convolution and depth-separable convolution is calculated by the following formulas:

$$C_{\text{std}} = k^2 * H * W * \text{In}_{\text{ch}} * \text{Out}_{\text{ch}} \tag{5}$$

$$C_{\text{sep}} = k^2 * \text{In}_{\text{ch}} * W * H + \text{In}_{\text{ch}} * \text{Out}_{\text{ch}} * W * H \tag{6}$$

where C_{std} denotes the standard convolution computation, C_{sep} represents the depth-separable convolution computation, H denotes the height of input feature map, W denotes the width of input feature map, In_{ch} indicates the number of input channels, Out_{ch} refers to the number of output channels, and k signifies the convolution kernel size.

Second, the inverted residual linear bottleneck structure proposed in MobileNetv2 [49] is introduced in the backbone layer, and the low-rank nature of the problem is adopted to improve the efficiency of the layer structure. Contrary to the residual structure, the inverse residual structure first enlarges the dimension of the input feature matrix by 1×1 convolution to increase the channel size. Then, convolution processing is performed in a higher dimensional space through 3×3 channel-by-channel convolution to capture more feature information. Finally, dimensionality reduction is performed by 1×1 convolution to compress the data by mapping the high-dimensional features to a low-dimensional space to reduce the parameters of the network. In high latitude space, activation functions such as ReLu can effectively increase the nonlinear expression of features, but only in high latitude space. In the inverted residual structure, the use of nonlinear activation results in greater feature loss because the number of channels is compressed at the end. With the purpose of preventing nonlinearity from destroying a large amount of feature information, a linear bottleneck is inserted after the last 1×1 convolution module of the inverted residual structure to replace the original nonlinear activation transformation, forming an inverted residual linear bottleneck structure to capture the region of interest. In terms of activation function, the inverted residual linear bottleneck structure uses the ReLU6 activation function, which is defined in the following formula:

$$\text{ReLU6}(x) = \min(\max(0, x), 6). \tag{7}$$

The activation function limits the maximum output to 6. Use this activation function for good performance on mobile devices with low precision. However, in the process of gradient descent, the intensity of descent completely depends on the product of weights, which may lead to gradient explosion with the reverse feedback of the algorithm. Swish nonlinear activation function [50], as an alternative to ReLU, can significantly improve the accuracy of the network, but its calculation and derivation are complex and unfriendly to the quantization process, especially for mobile devices. Therefore, combining the advantages of MobileNetV3, the relu6 activation function before the depth-separable convolution operation in the inverted residual linear bottleneck structure is replaced by the H_{swish} activation function [51], which is fit using the segmented linear function H_{Sigmoid} [52] to replace the sigmoid function in the swish activation function. It is defined as follows:

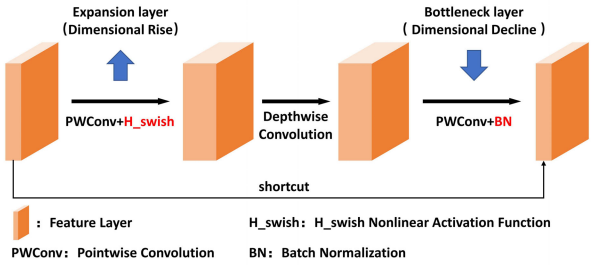


Fig. 4. Inverted residual linear bottleneck structure with the introduction of H_{swish} activation function.

$$H_{\text{swish}} = x * H_{\text{Sigmoid}} = x(\text{ReLU6}(x + 3))/6. \tag{8}$$

The H_{swish} activation function consists of a number of common operators. It reduces the number of memory accesses with little impact on performance, significantly reducing latency costs and making it more suitable for embedded devices. Compared to other activation functions, it is faster to compute and more friendly to quantization. Fig. 4 shows the inverted residual linear bottleneck structure with the introduction of H_{swish} activation function.

B. Feature Fusion Network Module Based on Attention Mechanism and Multiscale Feature Fusion

Although the network is lightweight, the amount of model calculation is considerably reduced. However, the lightweight network structure is too simplistic to fully extract the feature information and location information of the target to be detected. Moreover, remote sensing images have the problem of large target scale changes, many small targets, and difficult detection, resulting in a decline in detection accuracy. In this article, the CA mechanism [53] and the BiFPN structure [54] are introduced in the neck layer, and an efficient multiscale feature fusion network module based on the attention mechanism is proposed, so as to ensure the detection accuracy while reducing the weight. Compared with other attention mechanisms, the channel attention mechanism (squeeze-and-excitation) [55] ignores the importance of position information in the encoding process, and the convolutional block attention module (CBAM) [56] mechanism cannot capture long-distance dependencies. In addition, the following efforts are made to avoid compressing all spatial information into channels. First, the CA mechanism decomposes the global average pooling operation. Average pooling is performed on the input feature map with a size of $C \times H \times W$ in the horizontal and vertical directions to obtain feature maps in two directions of $C \times H \times 1$ and $C \times 1 \times W$. Second, concat is performed in the result on the spatial dimension. The channel is compressed by convolution, and then, split is separated. The number of channels of the two-direction feature vectors is readjusted by 1×1 convolution. Finally, the spatial information is fused by weighting the spatial dimension. The CA mechanism not only pays attention to channel information but also includes direction and position-sensitive information, avoiding the loss

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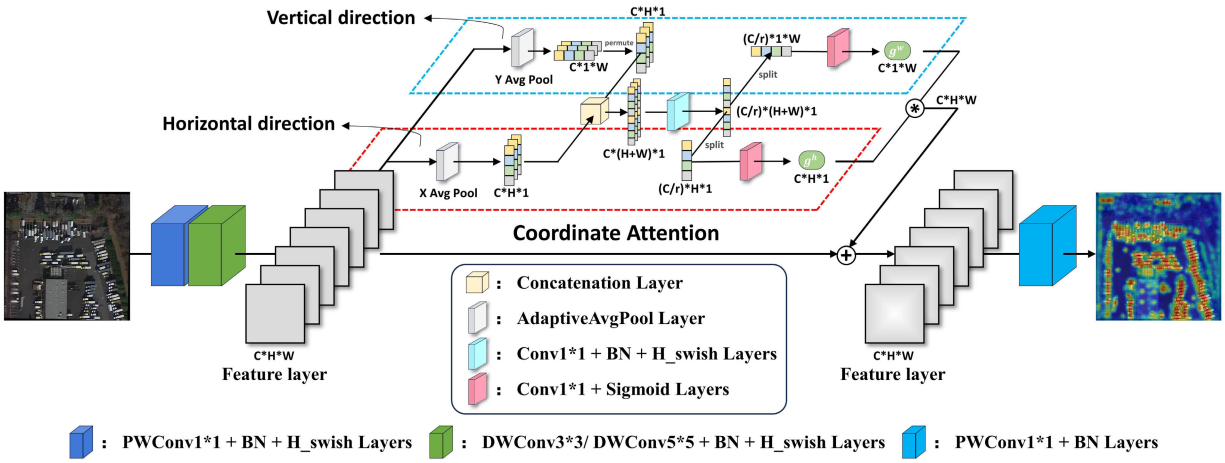


Fig. 5. Inverted residual linear bottleneck structure combined with CA mechanism.

of position information in 2-D global pooling while capturing long-range dependencies. Two strategies are employed in this study for improvement: 1) with respect to the inverted residual linear bottleneck structure in the backbone, a CA mechanism is introduced between channel-by-channel convolution and point-by-point convolution and 2) three CA modules of different dimensions are added separately before the convolution operation between the neck layer and the prediction layer, enabling the model to more accurately locate and identify target areas of different sizes for enhancing the feature representation of the lightweight network. Fig. 5 shows the module structure of the inverted residual linear bottleneck structure with the introduction of CA mechanism.

During feature fusion, the FPN [57] in the original structure of YOLO directly adds the features. Since different input features have different resolutions, their effects on the fused output features are often unequal. To wrestle with this difficulty, the BiFPN structure is introduced into the neck layer. The BiFPN structure was first applied in EfficientDet. Different from the previous feature fusion, the BiFPN structure adopts a multilevel feature fusion mechanism, uses horizontal and vertical information transmission to realize the comprehensive utilization of multiscale features, and adopts an adaptive feature weighting mechanism. This assigns different weights to each feature, allowing the network to better balance feature information at different scales and reduce unnecessary feature node connections. In the process of feature fusion in the neck layer, in accordance with the structural characteristics of BiFPN, the input and output nodes of the same layer across layers are connected, the information transmission path between the upper and lower layers is shortened. Learnable weights are added and set in the optimizer parameters: node weight ω_n and feature weighting coefficient ω_f. For each node, a convolutional layer is utilized to fuse multiple input feature maps, and then, a node weight is obtained through the sigmoid activation function to adjust the importance of each node in feature fusion. Concerning each feature map, a convolutional layer is adopted to map its feature vector to a scalar, and then, the sigmoid function is added to weight the features of each feature map for adjusting the contribution of different feature maps and enhancing the expressiveness of the model. The feature weighting coefficient is calculated by the following formula:

$$\omega_i = \sigma(f(\boldsymbol{F}_i)) \tag{9}$$

where $\boldsymbol{F}_i$ denotes the ith input feature diagram, f represents the convolution operation, σ indicates the sigmoid function, and ω_i refers to the feature weighting coefficient of the ith feature graph. We multiply all the feature maps by the corresponding weight coefficient, and Concat is the feature maps to get a new feature map, which is calculated by the following formula:

$$\boldsymbol{F}_{\text{out}} = \text{Concat}\left(\omega_0 * \boldsymbol{F}_0, \omega_1 * \boldsymbol{F}_1, \ldots, \omega_{(n-1)} * \boldsymbol{F}_{(n-1)}\right) \tag{10}$$

where Concat denotes the feature splicing function; $\boldsymbol{F}_{\text{out}}$ represents the output feature map; $\omega_0, \omega_1, \ldots, \omega_{(n-1)}$ indicate the feature weighting coefficients; and $\boldsymbol{F}_0, \boldsymbol{F}_1, \ldots, \boldsymbol{F}_{(n-1)}$ refer to the input feature maps. Fig. 6 shows the efficient feature fusion module proposed in this article.

C. Activation Functions

After finalizing our proposed AMFLW-YOLO network architecture, we further optimized the network in terms of activation function selection. The activation function provides nonlinearity to the network and gives the network more expressive power. In the original structure, the convolutional module in the neck layer uses the swish activation function, which applies sigmoid linear units by element and possesses the characteristics of no upper bound but the presence of a lower bound, smoothness, and nonmonotonicity, with obvious performance advantages in deeper networks. The expression for the swish activation function is calculated by the following formula:

$$f(x) = x * \sigma(x) \tag{11}$$

where σ denotes the sigmoid activation function.

However, the use of the sigmoid activation function in the network is prone to gradient disappearance, and the sigmoid

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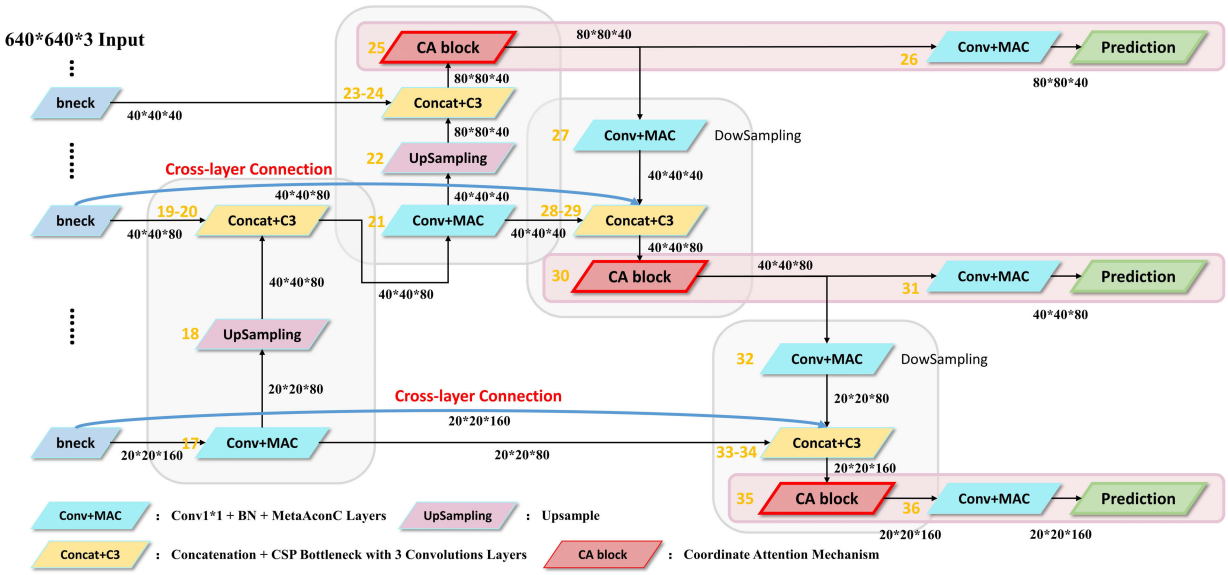


Fig. 6. Feature fusion module of AMFLW-YOLO network.

value often needs to be computed in power during the network training process, which results in a huge computational effort. In order to further lighten the network, we use the meta-AconC activation function [58] to replace the swish activation function. The expression for the swish activation function is calculated by the following formula:

$$f(x) = (p1 * x - p2 * x) * \text{sigmoid}(\beta * (p1 * x - p2 * x)) + p2 * x \quad (12)$$

where $p1 * x$ and $p2 * x$ denote the linear functions and β represents the switching factor. The design space of meta-AconC activation function includes layerwise, channelwise, and pixelwise, which correspond to layer, channel, and pixel, respectively. We adopt the channelwise scheme to perform adaptive learning for the whole channel by averaging the H and W dimensions separately. All pixels of each channel share a weight by two fully connected operations. The parameter β is derived from the sigmoid function. The linear and nonlinear states of the activation function are dynamically controlled using the parameter β, allowing the neurons to adapt themselves to the activation state. This initiative significantly reduces the network parameters and improves the performance significantly.

IV. Experimental Results and Discussion

In this section, the detection performance of the proposed model is evaluated on two public remote sensing datasets and a self-built dam remote sensing dataset, and the proposed method is compared with existing classical object detection algorithms to demonstrate the superiority of the proposed algorithm.

A. Datasets and Training

The RSOD dataset [59] is an optical remote sensing dataset publicly released by Wuhan University. It contains 976 high-resolution satellite images in four categories with a total of 6950 instances. Object classes consist of airplanes, playgrounds, overpasses, and oil barrels. The image resolution is 800 × 800 pixels.

The DOTA dataset [60] is a large-scale benchmark dataset for remote sensing object detection. The images in the DOTA dataset are primarily from Google Earth, and a small amount is from the JL-1 satellite and GF-2 satellite. It contains 2806 images, 15 categories, and a total of 188 282 instances. The resolution of the original images in the dataset ranges from 800 × 800 to 4000 × 4000 pixels.

The image resolution of the DOTA dataset ranges from 800 to 4000 pixels, and the aspect ratio of some images is too large, making it impossible to train directly. Therefore, the images in the dataset are cut into 600 × 600 fixed-size images, and the corresponding annotation information files are generated based on the images. Since cutting caused some images to have empty annotation targets or out-of-bounds annotation targets, the annotation information files and the corresponding images that do not meet the requirements are deleted, and the final experimental data are divided into a training set and a verification set by 4:1. The final experimental data contain 16 505 images in the training set and 4388 images in the verification set.

In order to further test the detection effect of the proposed algorithm, a new dam target detection dataset for remote sensing images (DTDRSI) is introduced. The 880 aerial dam images are collected from different sensors and platforms. Each image is 600 × 600 pixels in size. The dataset contains dam targets of various scales, orientations, and shapes. In order to avoid model overfitting due to the small amount of data, data augmentation is performed on the single sample. Specifically, the dataset is expanded using horizontal flip and random direction rotation operations, and the final obtained dataset is divided by 4:1 to obtain the training and validation sets. The final experimental data contain 2054 images in the training set and 586 images in the validation set. The dataset is labeled by labelImg, and the fully labeled DTDRSI dataset contains

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Fig. 7. Samples of annotated images in the DTDRSI dataset.

a total of 2759 instances. Some samples of annotated images in DTDRSI are shown in Fig. 7.

The AMFLW-YOLO network is evaluated on a 64-bit Windows 10 operating system using the Python programming language and the Pytorch library. The processor is Intel Core i7-7700HQ. The graphics card is NVIDIA Geforce GTX 1080 with 8-GB memory.

The loss function of the AMFLW-YOLO model consists of three parts: classification loss, confidence loss, and localization loss. For the small-size dense target, the classification loss and confidence loss are adopted as varifocal loss (VFLoss) [61], which uses the idea of sample weighting to solve the problem of positive and negative sample imbalance. The expression of the loss function is calculated by the following formula:

$$\mathrm{VFLoss}(p,q) = \begin{cases} -q(q\log(p)+(1-q)\log(1-p)), & q>0 \\ -\alpha p^{\gamma}\log(1-p), & q=0 \end{cases} \tag{13}$$

where p denotes the predicted perceptual classification score, q represents the target score, and α indicates the adjustable scaling factor. The positioning loss utilizes the complete intersection over union (CIoU) loss to calculate only the positioning loss of positive samples. The confidence losses of the three prediction feature layers employ different weights. Specifically, a weight of 4.0 is used in the prediction feature layer for predicting small objects, a weight of 1.0 is used for the prediction feature layer for predicting medium targets, and a weight of 0.4 is used in the prediction feature layer for predicting large objects. During the training process, the warm-up [62] and cosine annealing learning rate optimization strategy [63] are adopted. This strategy causes the learning rate to increase linearly from a very small value to a preset value to prevent overfitting of the model. Subsequent decay according to the cosine function values is performed to accelerate the model convergence. The optimizer weight decay and warm-up initial momentum are set to 0.0005 and 0.8, respectively; each batch contains four images; the input image resolution is 640 × 640 pixels; the models utilized in all experiments are trained on the training set for 300 epochs; and the initial learning rate is set to 0.0032.

B. Performance Evaluation of Model Lightweighting

Our method (AMFLW-YOLO) is compared with the benchmark model YOLOv5l and the existing advanced lightweight single-stage object detection algorithms, including Lite-YOLOv5 [64], YOLOv7-Tiny [65], YOLOX-s [66], YOLOX-Tiny, PicoDet-L [67], and YOLObile [68], regarding the model parameters, model size, floating-point operations (FLOPs), and detection speed. FLOPs are used to measure the computational complexity of the model.

TABLE I

MODEL PARAMETER QUANTITY COMPARISON. THE EXPERIMENTAL DATA WERE OBTAINED BASED ON THE RSOD DATASET

Methods	Params/M	Model size/MB	FLOPs/G	Speed/ms
YOLOv5l	47.06	103.1	111.4	8.9
Lite-YOLOv5	5.3	10.9	15.7	6.9
YOLOv7-Tiny	6.01	26.4	13.2	6.5
YOLOX-s	8.94	35.2	26.8	7.1
YOLOX-Tiny	5.06	10.2	6.45	6.1
PicoDet-L	3.3	8.9	8.91	5.8
YOLObile	4.59	9.7	8.76	6.4
AMFLW-YOLO (Ours)	1.97	4.4	4.62	5.6

TABLE II

RESULTS OF ABLATION EXPERIMENT ON THE RSOD DATASET

Methods	DS	IS(H_S)	CA	BiFPN	MA	mAP/%	Params/M
YOLOv5l						92.5	47.06
methods(1)	✓					86.3 (-6.2)	4.27 (-42.79)
methods(2)	✓	✓				88.6 (-3.9)	3.51 (-43.55)
methods(3)	✓	✓			✓	90.3 (-2.2)	3.31 (-43.75)
methods(4)	✓	✓	✓		✓	93.7 (+1.2)	3.78 (-43.28)
methods(5)	✓	✓	✓	✓	✓	**95.0 (+2.5)**	**1.97 (-45.09)**

Table I lists the comparison results between the proposed method and the abovementioned state-of-the-art methods in model parameters, model size, FLOPs, and detection speed. AMFLW-YOLO has the smallest number of parameters and model size (params: 1.97 M and model size: 4.4 MB). The model parameters are 40.3% lower than the minimal algorithm (PicoDet-L). The number of parameters is curtailed by 97.4% compared with the original structure YOLOv5l benchmark. The size of the model trained using the RSOD dataset is only 4.4 MB, which is 95.7% smaller than the original structure YOLOv5l benchmark. It is worth noticing that our model has lower complexity than other networks. AMFLW-YOLO has 4.62-G FLOPs (multiply–adds), which is only 4% of YOLOv5l (111.4G FLOPs). The experimental results reveal the effectiveness of the model lightweight method in this article. AMFLW-YOLO is more suitable for application in microembedded devices. Simultaneously, the detection speed of AMFLW-YOLO is 5.6 ms/frame, which satisfies the real-time detection requirements of remote sensing targets.

C. Ablation Experiment

To assess the detection performance of our proposed AMFLW-YOLO algorithm, we conduct ablation experiments to examine the impact of each improvement made to the algorithm (model lightweighting, attention mechanism, and multiscale feature fusion) on its performance. The performance evaluation metrics include the mean average precision (mAP) and params.

Table II lists the experimental results obtained by AMFLW-YOLO for the RSOD dataset under various optimization measures. "DS," "IS(H_S)," and "MA" stand for depth-separable convolution, inverted residual linear bottleneck combined with H_{swish} activation function, and meta-AconC

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TABLE III
DETAILED RESULTS FOR FOUR CATEGORIES OF RSOD DATASET

Class	YOLOv5l	Lite-YOLOv5	YOLOv7-tiny	YOLOX-s	YOLOX-tiny	PicoDet-L	YOLObile	AMFLW-YOLO
aircraft	94.2	91.1	93.8	94.2	89.8	96.3	90.0	97.4
playground	98.4	95.4	98.9	98.3	93.9	99.0	92.7	99.5
overpass	81.0	75.2	72.7	78.1	70.2	83.9	71.1	85.0
oiltank	97.1	94.0	96.9	96.1	92.9	96.8	90.2	97.9
mAP	92.3	88.1	90.6	91.8	83.6	93.8	83.2	95.0
PSD	6.92	8.12	10.48	7.97	10.20	5.92	9.29	5.80

TABLE IV
DETAILED RESULTS FOR 15 CATEGORIES OF DOTA DATASET

Class	YOLOv5l	Lite-YOLOv5	YOLOv7-tiny	YOLOx-s	YOLOX-tiny	PicoDet-L	YOLObile	AMFLW-YOLO
plane	96.8	94.9	95.8	94.7	89.7	97.0	89.2	97.2
baseball-diamond	93.4	83.8	88.1	88.9	80.4	92.1	78.9	93.6
bridge	67.1	49.0	53.6	50.4	46.2	67.8	45.8	69.3
ground-track-field	78.9	62.1	62.0	64.1	61.6	73.5	63.0	75.2
small-vehicle	75.4	65.8	60.1	62.6	64.3	76.6	61.7	80.5
large-vehicle	96.8	88.3	90.3	88.2	88.0	94.7	86.9	95.7
ship	94.5	92.9	80.1	93.7	87.6	96.2	86.0	96.1
tennis-court	96.0	92.8	91.2	92.0	88.9	97.0	89.1	97.9
basketball-court	60.7	58.1	58.9	60.7	53.0	63.1	51.9	67.4
storage-tank	85.4	72.7	73.6	70.2	72.5	82.8	71.7	84.6
soccer-ball-field	72.0	62.0	63.8	67.1	60.3	73.6	61.5	74.7
roundabout	70.6	66.8	66.0	69.5	64.5	72.9	64.3	74.2
harbor	91.2	85.3	84.1	75.0	82.0	91.0	79.6	90.1
swimming-pool	80.3	70.0	72.9	72.7	72.1	81.5	71.0	83.8
helicopter	47.3	42.6	41.9	50.6	40.3	49.2	40.7	50.0
mAP	83.4	75.2	76.1	76.9	72.0	84.0	71.2	84.7
PSD	14.78	16.20	15.79	14.92	15.62	14.22	15.46	13.53

activation function. Method (1) shows that when we lighten the network by replacing the standard convolution with depth-separable convolution alone, the number of model parameters is reduced geometrically and substantially, but it also causes a loss in detection accuracy. The mAP and params of the network are reduced by 6.2% and 42.79 M, respectively. Methods (2) and (3) show that when the inverted residual linear bottleneck structure is used for the feature extraction layer and the SiLu activation function is also replaced with the meta-AconC activation function, the number of parameters in the network is further reduced based on the previous one. The mAP is improved by 2.3% and 1.7% compared with the previous step but still lower than the baseline value. It is proved that the inverted residual linear bottleneck structure and meta-AconC activation function have performance advantages in lightweight networks. Method (4) shows that when we introduce the CA mechanism on top of the lightweighting, we can improve the mAP by 3.4% based on only increasing the number of parameters by 0.47 M. Method (5) shows that the introduction of BiFPN structure on the previous basis improves the mAP by 1.3% over the previous step, while the network size is further compressed. The model structure of AMFLW-YOLO selects Method (5) to ensure the optimal model performance. AMFLW-YOLO finally obtains 95% mAP on the RSOD dataset, and params is 1.97 M. Compared with the YOLOv5l baseline, the mAP is improved by 2.5%, and the params is reduced by 45.09 M. We use Grad-CAM to perform interpretability analysis of the model improvement strategy, and Fig. 8 shows the heat maps obtained by Grad-CAM for YOLOv5l and AMFLW-YOLO on RSOD and DOTA datasets for small-size targets and large-size targets, and it can be observed that compared with YOLOv5l, AMFLW-YOLO has a higher detection target location region more attention to the nontarget region and less attention to the irrelevant environmental information for the nontarget region. Also, for different sizes of targets, AMFLW-YOLO can better focus attention on positive sample regions while reducing the attention on irrelevant information in the environment. This indicates that the introduction of CA mechanism based on lightweight network enhances the feature representation of the network, while the adoption of BiFPN structure significantly improves the detection of the model for remote sensing targets of different scales. The experimental results verify the effectiveness of each improvement measure.

D. Comparison With State-of-the-Art Methods

We compare the detection performance of the method in this article (AMFLW-YOLO) with that of the benchmark model YOLOv5l and existing state-of-the-art lightweight single-stage target detection algorithms, including Lite-YOLOv5, YOLOv7-Tiny, YOLOX-s, YOLOX-Tiny, PicoDet-L, and YOLObile. The performance evaluation metrics include the average precision (AP) of each class, the mAP of all classes, and the population standard deviation (PSD). The AP and mAP are used to evaluate the object detection performance. We follow the PASCAL VOC2007 benchmark to calculate the AP and mAP, and the IoU threshold is 0.5. Table III shows the experimental results obtained by the above advanced methods on the RSOD dataset, and the experimental results show that our method obtains 95% of the mAP, which is a small improvement in detection accuracy compared to the YOLOv5 benchmark network. Compared with Lite-YOLOv5, the mAP of our method increased by 6.9%. Table IV shows

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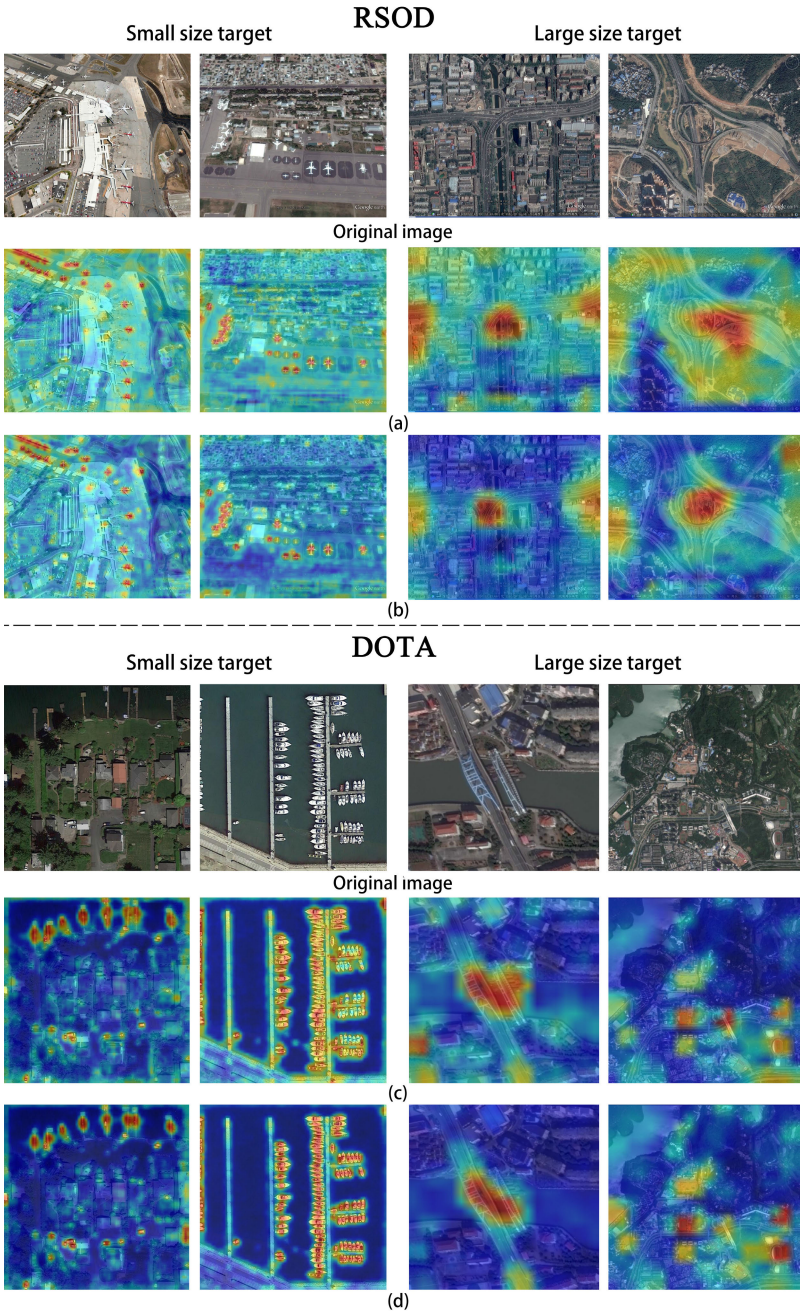


Fig. 8. Heat maps of YOLOv5l and AMFLW-YOLO for small-size targets and large-size targets obtained by Grad-CAM. (a) Heat maps generated by YOLOv5l for small and large targets on the RSOD dataset. (b) Heat maps generated by AMFLW-YOLO for small and large targets on the RSOD dataset. (c) Heat maps generated by YOLOv5l for small and large targets on the DOTA dataset. (d) Heat maps generated by AMFLW-YOLO for small and large targets on the DOTA dataset.

the experimental results obtained by the above advanced methods on the DOTA dataset, and our method obtains 84.7% of the mAP. Compared with Lite-YOLOv5, YOLOv7-tiny, YOLOX-tiny, PicoDet-L, and YOLObile, the overall performance of AMFLW-YOLO is greatly improved, especially for small objects. For example, the AP of AMFLW-YOLO is 80.5% in the small-vehicle category, and it increased by 20.4% compared with YOLOv7-tiny and 18.8% compared with YOLObile. Table IV also shows that the "plane," "large-vehicle," "ship," "tennis-court," and "baseball-diamond" categories reach better detection performance. This is probably because these object categories have smaller in-category variations.

PSD is used to measure the extent to which the AP of single class deviates from the mAP of all classes to reflect the stability of the model's detection performance. Table III shows that the PSD of the AMFLW-YOLO detection results is 5.80. Table IV shows that the PSD of the AMFLW-YOLO detection results is 13.53. The comparison of experimental results shows that AMFLW-YOLO has the smallest discretization of detection results on the two datasets, and the model detection performance is more stable.

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TABLE V
DETAILED RESULTS FOR ONE CATEGORY OF DTDRSI DATASET

Class	YOLOv5l	Lite-YOLOv5	YOLOv7-tiny	YOLOx-s	YOLOX-tiny	PicoDet-L	YOLObile	AMFLW-YOLO
dam	88.1	83.0	85.4	86.0	85.7	89.7	81.6	90.9
mAP	88.1	83.0	85.4	86.0	85.7	89.7	81.6	90.9

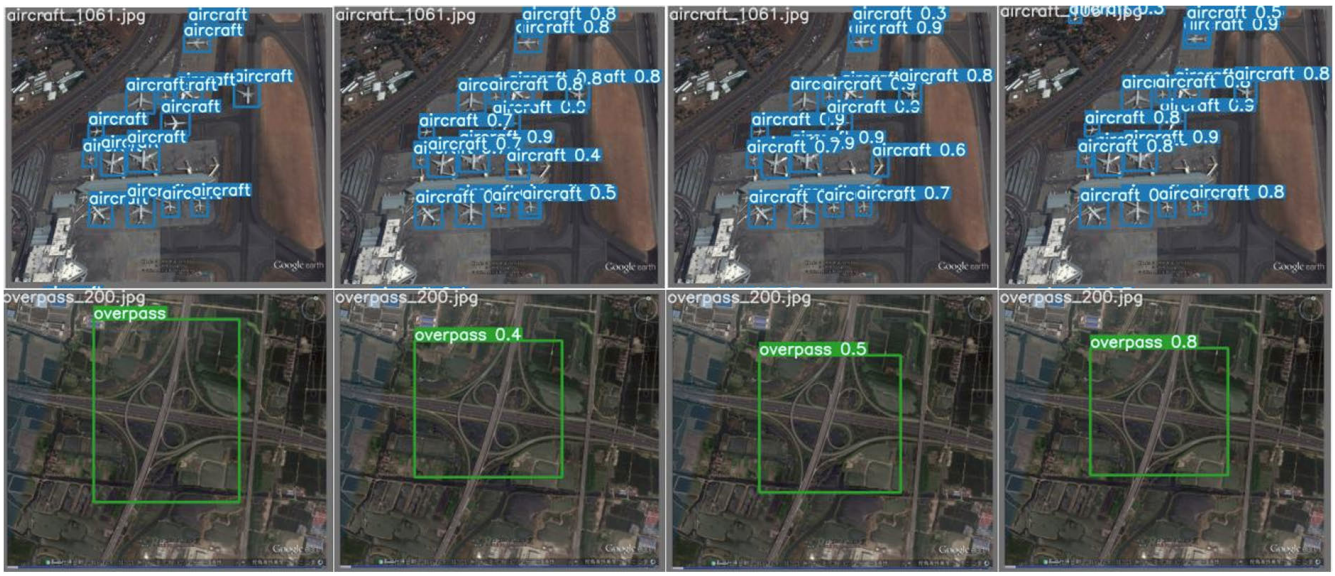


Fig. 9. Sample detection results from YOLOv7-tiny, YOLOv5l, and AMFLW-YOLO. (a) Labels. (b) YOLOv7-tiny. (c) YOLOv5l. (d) AMFLW-YOLO.

Table V shows the experimental results obtained by the above advanced methods on the DTDRSI dataset, and our method obtains 90.9% of the mAP. In order to highlight the superiority of the proposed method, the experimental results of Lite-YOLOv5 and YOLOX-tiny on the DTDRSI dataset are provided. The experimental results show that both Lite-YOLOv5 and YOLOX-tiny obtain a small decrease in mAP on the test set compared to YOLOv5l and YOLOx-s, while AMFLW-YOLO gets 2.8% and 4.9% rise in mAP on the test set. It indicates that the method proposed in this article can better maintain the detection performance based on network lightweighting.

As shown in Fig. 9, we select YOLOv7-tiny and YOLOv5l to compare the sample detection results with AMFLW-YOLO. Based on the sample detection results, we notice that both YOLOv7-tiny and YOLOv5 have different degrees of false detection for small-sized targets in remote sensing images, and our method reduces the false detection rate significantly while improving the detection accuracy. The improved performance is mainly due to the CA mechanism and BiFPN structure adopted in this article, which better balance the feature information at different scales and reduce the possibility of pseudo-target interference. Fig. 10 shows the detection effect of AMFLW-YOLO on the RSOD and DOTA datasets. Fig. 11 shows the detection effect of AMFLW-YOLO on the DTDRSI dataset.

In short, the overall experimental results show that for remote sensing targets in remote sensing images with complex backgrounds, our proposed AMFLW-YOLO algorithm has a better performance than the above target detection algorithms in terms of model lightweighting and detection accuracy.

E. Discussion

The algorithm proposed in this article is based on the idea of multiscale and utilizes an efficient bidirectional cross-scale connectivity and weighted feature fusion structure. The high-level features are enhanced by combining multiple low-level features to increase the feature dimensions and information. For small target detection in remote sensing images, the algorithm proposed in this article performs better than the above algorithms. However, small targets occupy fewer pixels in the image and have lower resolution. Therefore, there is no guarantee that the constructed features are sufficiently interpretable and discriminative for small targets. The supervision of small targets during the training process is not perfect enough, and the loss of small targets contributes less to the loss of the overall model, which leads to the effect of the extracted features that is still not very good. With the gradual application of deep learning in super-resolution technology, the small target detection methods based on super-resolution are relatively explanatory and have very impressive results. However, these algorithms require relatively high hardware requirements and rely heavily on massive amounts of data. In future work, feature enhancement for small-target detection using super-resolution techniques will be the focus of our research. We will also consider specifically focusing on small-target resampling during the training process to improve

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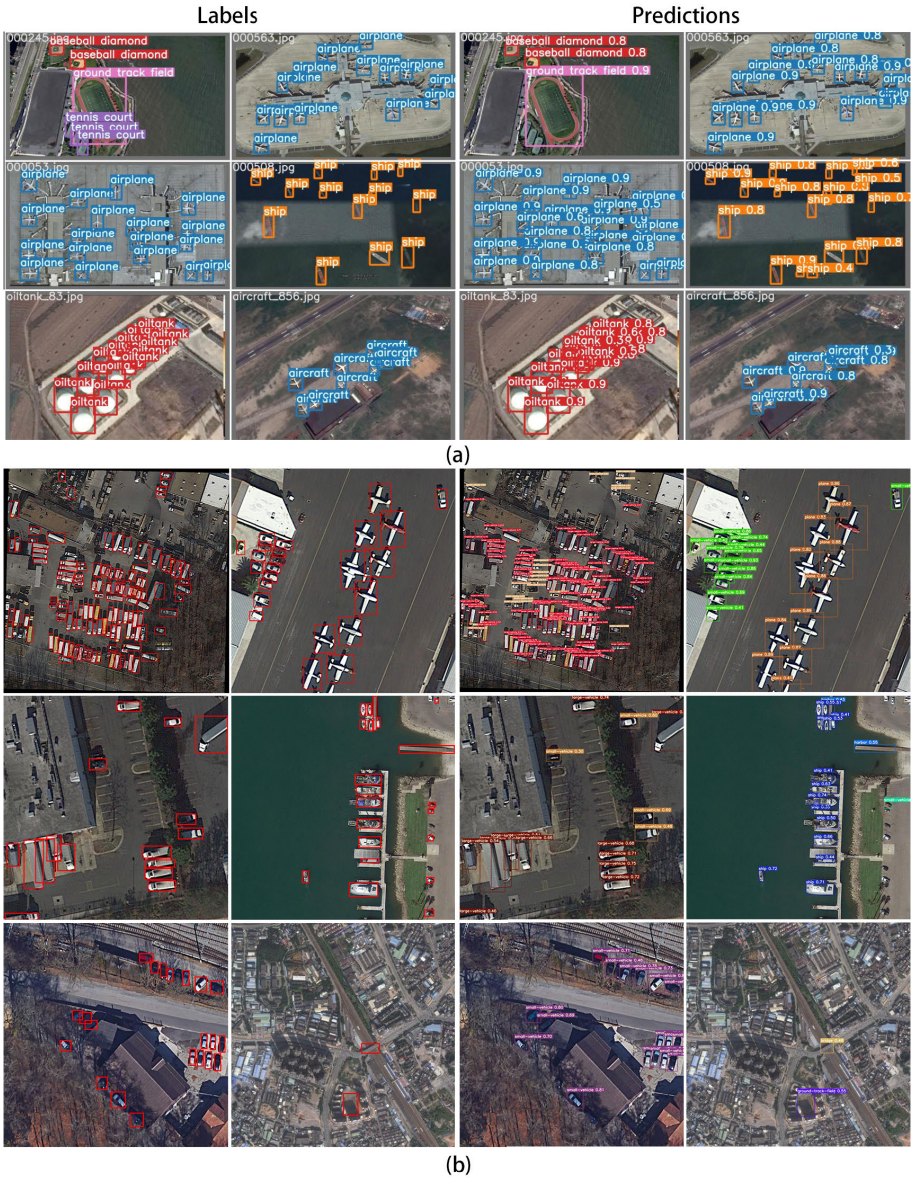


Fig. 10. Detection effect of AMFLW-YOLO on (a) RSOD and (b) DOTA datasets.

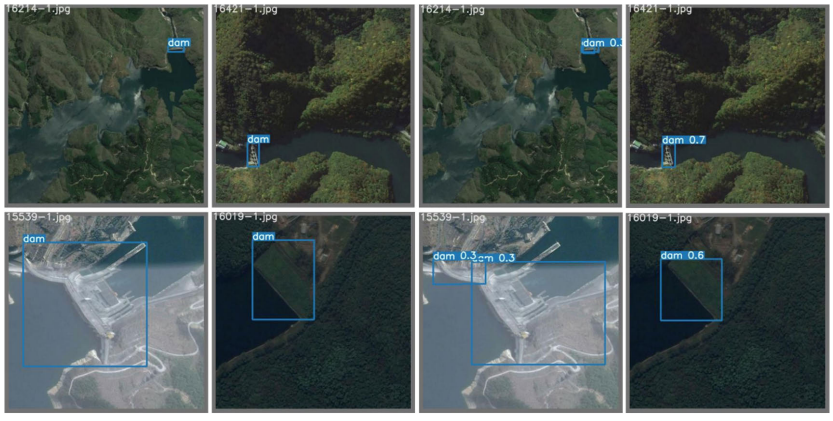


Fig. 11. Detection effect of AMFLW-YOLO on the DTDRSI dataset.

the quality of training. In addition, the object detector proposed in this article is designed for inference of independent images and achieves good results, but it lacks the correlation between multiple frames. The use of the proposed AMFLW-YOLO

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combined with the spatiotemporal relationship between multiframe images for remote sensing image change detection will be the focus of our future research.

V. Conclusion

A lightweight remote sensing image detection algorithm AMFLW-YOLO based on attention mechanism and multiscale feature fusion is proposed in this article. The algorithm combines lightweight network modules, attention mechanism, and multiscale feature fusion to balance the lightweight network and algorithm accuracy. First, deep separation convolution and antiresidual linear bottleneck structure are employed in the feature extraction layer to significantly reduce the model parameters and achieve the lightweight of the model. Second, the CA mechanism is introduced in the feature fusion layer to enhance the feature representation in the network. An efficient bidirectional cross-scale feature fusion network is designed by referring to the BiFPN structure, which enhances feature extraction, better captures multiscale features, and effectively improves the detection accuracy of remote sensing targets in complex backgrounds. The experimental results on DOTA, RSOD, and DTDRSI datasets show that the method has good detection performance for remote sensing targets in complex backgrounds. In the future, we will explore methods to address spectral variability based on this article. Feature enhancement of features for small target detection using super-resolution techniques and remote sensing image change detection will also be the focus of our research.

Conflict of Interest

The authors declare that they have no financial and personal relationships with other people or organizations that can inappropriately influence in this article. There is no professional or other personal interest of any nature or kind in any product. They confirm that there is no conflict of interest regarding the publication in this article and declare that there is no conflict of interest regarding the publication of this article.

Data Availability

In this work, the open remote sensing datasets that support the findings of this study are available from GitHub. The DOTA dataset is available at http://captain.whu.edu.cn/DOTA web or https://captain-whu.github.io/DOTA. Published paper on DOTA dataset is available at https://arxiv.org/abs/1711.10398. The RSOD dataset is available at https://github.com/RSIA-LIESMARS-WHU/RSOD-Dataset-. Published paper on RSOD dataset is available at https://ieeexplore.ieee.org/abstract/document/7827088.

Author's Contributions

Dr. Guili Peng is the main contributor to this work. Zijian Yang has mainly contributed to modify of the algorithm. Prof. Shoubin Wang participated in the preparation of the article structure as a communicated author. Dr. Yuan Zhou has given guidance for this article.

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Guili Peng was born in Tianjin, China, in 1981. He received the Ph.D. degree from the Southwest University of Science and Technology, Mianyang, Sichuan, China, in 2021.

He is currently a Lecturer with Tianjin Chengjian University, Tianjin. His research interests include geophysics, instrument science and technology, automatic control theory, electrical and electronic technology, microseismic data signal processing, and nuclear electronic technology.



Zijian Yang was born in Hebei, China, in 1999. He received the B.Eng. degree from the Liaoning University of Technology, Jinzhou, Liaoning, China, in 2021. He is currently pursuing the master's degree with Tianjin Chengjian University, Tianjin, China.

His research interests include deep learning, computer vision, remote sensing image detection, and intelligent construction technology.



Shoubin Wang was born in Henan, China, in 1979. He received the Ph.D. degree from the Harbin Institute of Technology, Harbin, Heilongjiang, China, in 2015.

He is currently a Professor with Tianjin Chengjian University, Tianjin, China. His research interests include automatic detection and control technology, artificial intelligence, smart city, temperature inversion, automatic control, and artificial intelligence.



Yuan Zhou was born in Nanyang, Henan, China, in 1993. He received the master's degree from the Hebei University of Technology, Tianjin, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Instrument Science and Engineering, Harbin Institute of Technology, Harbin, China.

His research interests include event-triggered control theory, noncontact measurement, inverse problems, and machine vision.

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